

# SOHAM ROY

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## EDUCATION

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### Birla Institute of Technology and Sciences(BITS), Pilani

Bachelor of Computer Science Engineering

Expected May 2028

**Relevant Coursework:** DSA, DBMS, AI, Digital Design, Computer Architecture., OS, Compiler Construction, Computer Networks

### 12th Grade-86% , 10th Grade-90%

Secured an All India Rank of **6860** in the IIT-JEE examination

Secured **98.32 %ile** in the JEE Mains examination

## SKILLS

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**Languages and Databases::** C++, C#, Java, Python, JavaScript, TypeScript, CSS, SQL, Verilog, Bash, MongoDB.

**AI and Backend::** Pytorch, TensorFlow, NodeJS, React, Langchain, ChromaDB, FAISS, Docker, Kubernetes, RabbitMQ,

**Developer Tools and Systems:** Git, GitHub, COMSOL, Flutter, Grafana, NetCat, WireShark, Postman.

**Certifications & Training:** Cisco Networking Academy: Introduction to Cybersecurity

Cisco Networking Academy: Ethical Hacking

Attended the accredited Indian Workshop on Quantum Programming by QCIG

German A1 Grade Exam: 51/60

Prepping for CompTIA Sec+

## WORK EXPERIENCE

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### CyberArk & BITS Pilani(Under Dr. Ashutosh Bhatiya)

Research Assistant

October 2025 – January 2026

- Collaborated on a **Zero-Trust Architecture** framework for securing LLM-based agents.
- Designed policies for safe tool invocation and auditing to mitigate prompt- and tool-injection attacks and worked on a paper.

### Remote(Under Dr. Suresh Kumar)

Junior Researcher

September 2025 – December 2025

- Assisted in numerical analysis of cosmic background data; wrote scripts for **data preprocessing** and **spectral visualization**.
- Reviewed various research papers to understand **Cosmic Background Radiation**.

**FIDE(Beckn Protocol)** (building decentralized and interoperable P2P networks.)

Software Engineering Intern(Open Source Software)

May 2025 – July 2025

- Containerized **MCP**-based **microservices** on various API endpoints (Node.js, FastAPI) for **P2P** LLM trade orchestration.
- Optimized **RAG pipeline** reducing token usage by **32%**.

### BITS Pilani(University-Under Dr. Radha Raman Mishra)

Research Assistant

November 2023 – May 2024

- Simulated and tested **3D-printed magnetic PLA** structures under **microwave exposure** for **adaptive space components**.
- Used **COMSOL Multiphysics** for **modeling and correlation** with lab data.

## UNIVERSITY PROJECTS

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### Q-Learning in a 3D Gridworld-Reinforcement Learning

Oct 2025 – Nov 2025

- Played around with **reinforcement learning**, specifically the **Q-Learning formula** and its effect on a **Hidden Markov Model** to generate **graphs** and **heat-maps** using **Matplotlib** with different test cases.

### AI-Agent for Unstructured Data Analysis(Deep Researcher)

Jan 2025 – May 2025

- Developed an AI assistant integrating **web scraping**, **PDF parsing** and **semantic search** to answer queries based on user uploaded research papers using **RAG** via **Langchain** and **ChromaDB**
- Benchmarked on **customized and BERT-based rubric**, achieving 90% similarity with top commercial LLMs (GPT-3.5, Llama 2).

### Lexical Analyzer and Parser for a custom compiler

Jan 2025 – May 2025

- Designed and implemented a **DFA** for a **lexical analyzer** and **parser** in C based on requirements and test cases provided by a professor.
- Uses a **twin-buffer I/O** to skip repeated file reads and builds a **LL(1) parsing table** for error and sync issues.

## LEADERSHIP ACTIVITIES

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### BITS Coding Club: Game Development Vertical

#### Senior member

January 2024 – Present

- Built a **VR Game** from scratch. I worked primarily on **Blender** and Unity, and I expertise in **physics**, **animations** and **game design**.

### Matrix: Film & Literature Society

#### Vice President

August 2025 – Present

- Co-led the club on campus through 3 festivals involving public book-fests and movie-screenings.

### BITS Martial Arts

#### Team Member - 2 time medalist

August 2023 – May 2025

## LANGUAGES

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English(Fluent), Hindi(Fluent), Marathi(Fluent), Bengali(mother-tongue), German(Intermediate)

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